15H1. To prepare for the first model, lead the data and take a look at each variable:

R code 15.19
```
library(rethinking)
data(elephants)
d <- elephants
str(d)
```

```
'data.frame': 41 obs. of  2 variables:
 $ AGE     : int  27 28 28 28 28 29 29 29 29 29 ...
 $ MATINGS: int  0 1 1 1 3 0 0 0 2 2 ...
```

This is a very simple set of data. `AGE` contains ages in years of individual male elephants. `MATINGS` contains counts of matings for those individuals.

Before we jump into the model, let's think about how age should influence the number of matings. You could enter it as it is, in which case you are assuming an exponential relationship between age and matings, because the log link makes it so:

$$\log \lambda_i = \alpha + \beta A_i \implies \lambda_i = \exp(\alpha + \beta A_i) = \exp(\alpha) \exp(\beta A_i)$$

Maybe there is an exponential relationship with age. But there are other options. For example it could be that matings scale with the logarithm of age. Like this:

$$\log \lambda_i = \alpha + \beta \log A_i \implies \lambda_i = \exp(\alpha) A_i^{\beta}$$

This allows age to have increasing or diminishing returns on matings, depending up whether β is less than or greater than 1.

Let's use the logarithm approach. But using age directly as in a conventional GLM isn't wrong. Unlike in most examples, I'll keep age on the natural scale. This means that the intercept is now the expected matings at the start of sexual maturity (about 20 years old). So let's convert age to years after that by subtracting 20. The implied Poisson model predicting `MATINGS` with `AGE` is:

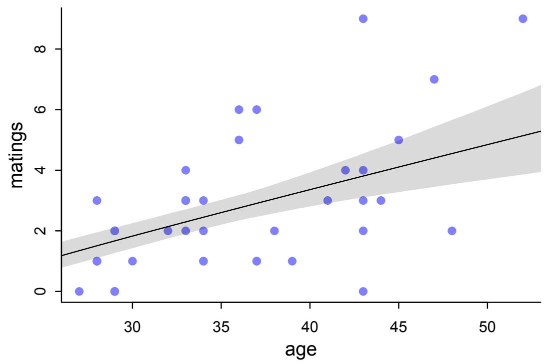
R code 15.20
```
m15H1.1 <- ulam(
    alist(
        MATINGS ~ dpois(lambda),
        lambda <- exp(a)*(AGE-20)^bA,
        a ~ dnorm(0,1),
        bA ~ dnorm(0,1)
    ), data=d , chains=4 )
precis( m15H1.1 )
```

```
    mean   sd  5.5% 94.5% n_eff Rhat4
a  -1.45 0.59 -2.41 -0.53   276  1.01
bA  0.89 0.21  0.55  1.22   270  1.01
```

These chains are not very efficient, but they did converge. Take a look at the trace plot, too. Let's plot the implied relationship:

R code 15.21
```
A_seq <- seq( from=25 , to=55 , length.out=30 )
lambda <- link( m15H1.1 , data=list(AGE=A_seq) )
lambda_mu <- apply(lambda,2,mean)
lambda_PI <- apply(lambda,2,PI)
```

```
plot( d$AGE , d$MATINGS , pch=16 , col=rangi2 ,
    xlab="age" , ylab="matings" )
lines( A_seq , lambda_mu )
shade( lambda_PI , A_seq )
```



That looks like a reliably positive relationship. Older elephants get more matings, on average.

Now let's assume each AGE value was measured with Gaussian error with standard deviation 5, as the problem suggests. Here's the new model. The only trick to observe, as in the chapter, is to manually add the [i] index to the predictor. I'm going to subtract 20 from the ages first, to make the coding simpler. We can add it back later, after sampling.

R code 15.22

```
d$AGE0 <- d$AGE - 20
m15H1.2 <- ulam(
    alist(
        MATINGS ~ dpois(lambda),
        lambda <- exp(a)*AGE_est[i]^bA,
        AGE0 ~ dnorm( AGE_est , 5 ),
        vector[41]:AGE_est ~ dunif( 0 , 50 ),
        a ~ dnorm(0,1),
        bA ~ dnorm(0,1)
    ), data=d , chains=4 )
precis( m15H1.2 )
```

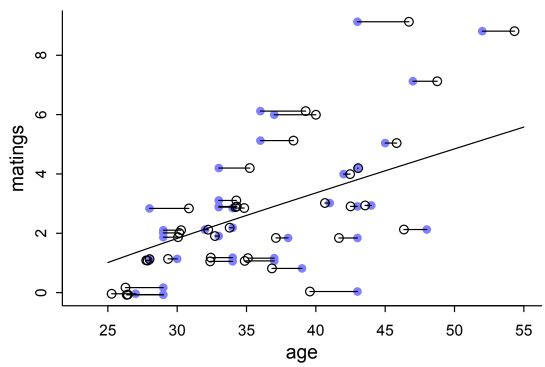
```
41 vector or matrix parameters hidden. Use depth=2 to show them.
    mean   sd  5.5% 94.5% n_eff Rhat4
a  -1.27 0.55 -2.18 -0.42   770     1
bA  0.83 0.19  0.53  1.13   808     1
```

Ignoring the AGE_est parameters for the moment, the average association between MATINGS and AGE has not changed. Why not? The measurement error is both symmetric and the same for all ages. So this is unlike the divorce rate example in the chapter in the sense that the error is uniform. So adding equal error to all of the predictor values doesn't have much impact. Here, it has had essentially no impact on inference. Chances are that real measurement error would not be uniform across all ages. It would be much easier to distinguish among young ages than older ages, for example.

We can learn something more in this example by inspecting the posterior distributions of the AGE values, the AGE_est parameters. Let's extract them and compare the posterior means to the observed values. The plot I'll construct will have AGE on the horizontal and MATINGS on the vertical, with open points for the inferred posterior means and filled blue points for the observed, as usual. I'll connect each pair of points for the same animal with a line segment. Since so many points overlap, I'll also add a little jitter to the vertical scale, so we can tell individuals apart more easily. Finally, I'll plot the mean regression trend.

R code 15.23

```
post <- extract.samples(m15H1.2)
AGE_est <- apply(post$AGE_est,2,mean) + 20

# make jittered MATINGS variable
MATINGS_j <- jitter(d$MATINGS)

# observed
plot( d$AGE , MATINGS_j , pch=16 , col=rangi2 ,
    xlab="age" , ylab="matings" , xlim=c(23,55) )

# posterior means
points( AGE_est , MATINGS_j )

# line segments
for ( i in 1:nrow(d) )
    lines( c(d$AGE[i],AGE_est[i]) , rep(MATINGS_j[i],2) )

# regression trend - computed earlier
lines( A_seq , lambda_mu )
```



The key thing to notice here is that the observations above the regression trend have been adjusted upwards in the posterior distribution, while the observations below the trend have been adjusted downwards. Why? Consider an observed number of matings that is above the expectation for a given observed age. This puts the blue point above the regression trend. For this observed age, measured with error, the matings exceed what is expected for that age. So the model "realizes" that the actual age of that individual is probably above what was measured and entered into the data table. So the open points (inferred) above the trend are to the right of the blue points (measured). Likewise, a blue point below the regression trend has a number of matings below what is expected for the measured age. So the model realizes that the actual age is probably below the measured age. So the open points (inferred) below the trend are to the left of the blue points (measured). Notice also that blue points farther from the regression trend shrink farther towards it. This is pooling, as usual.

15H2. All that's required to fit the new models is to adjust the `5` inside the distribution assigned to `AGE`. For example, let's begin by doubling the standard error to 10.

R code 15.24

```
m15H2.1 <- ulam(
    alist(
        MATINGS ~ dpois(lambda),
        lambda <- exp(a)*AGE_est[i]^bA,
        AGE0 ~ dnorm( AGE_est , 10 ),
        vector[41]:AGE_est ~ dunif( 0 , 50 ),
        a ~ dnorm(0,1),
        bA ~ dnorm(0,1)
    ), data=d , chains=4 )
precis( m15H2.1 )
```

```
    mean   sd  5.5% 94.5% n_eff Rhat4
a  -1.12 0.54 -2.01 -0.30   635  1.01
bA  0.75 0.18  0.49  1.05   665  1.01
```

Smaller, but not close to zero. Double again and we get:

```
    mean   sd  5.5% 94.5% n_eff Rhat4
a  -0.93 0.57 -1.90 -0.04   717  1.01
bA  0.65 0.18  0.37  0.94   705  1.01
```

At this rate, it might take a while. Let's jump to a standard deviation of 100:

```
    mean   sd  5.5% 94.5% n_eff Rhat4
a  -0.46 1.06 -1.76  1.92   170  1.02
bA  0.46 0.35 -0.36  0.86   161  1.02
```

Still not zero, and the chains sample very badly. But there is a lot of mass below zero for `bA`. The basic lesson is that as the error grows, it is increasingly hard to detect any association between age and mating.